RAMCO INSTITUTE OF TECHNOLOGY
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Department of Computer Science and Business Systems
Academic Year 2023 – 2024(Odd Semester)

Degree, Semester & Branch: B.Tech., I & CSBS
Course Code & Title: GE3151 Problem Solving and Python Programming
Name of the Faculty member: Ms.K.Usharani AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 5 / Modules, Packages
- **Course Outcome:** CO 5 – Illustrate read and write data from/to files in Python programs.
- **Topic Learning Outcome:** TLO 15 - Students will be able to utilize python modules, packages to develop solution for a problem
- **Activity Chosen:** Minute paper
- **Justification:**
    1. Minute paper activity will enable the students to recall the topics on their own.
    2. Better tool for brainstorming the ideas and concepts discussed in lecture.
    3. At the end of discussion on modules and packages, it is a better way to recall those concepts.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
    1. The students were asked to take a paper and pen.
    2. They were asked to the write the concepts of modules and packages individually.
    3. The students were asked to submit the paper.
    4. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

- **CO – PO / PSO mapping:**

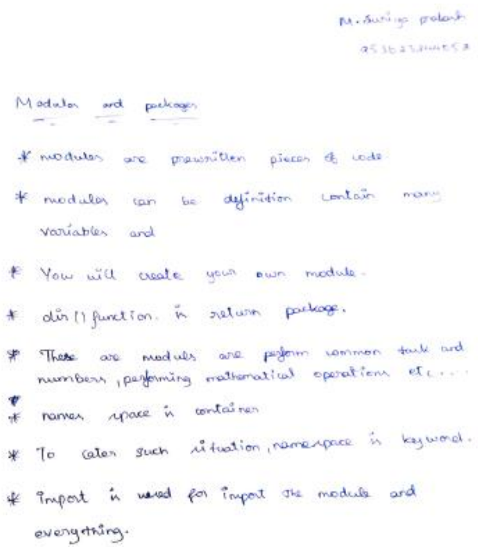
CO	PO1	PO2	PO3	PO9	PO12	PSO2
CO5	3	1	1	1	2	1

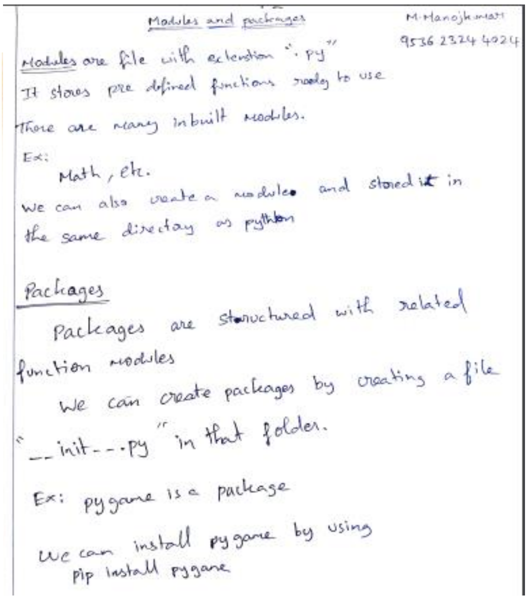
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

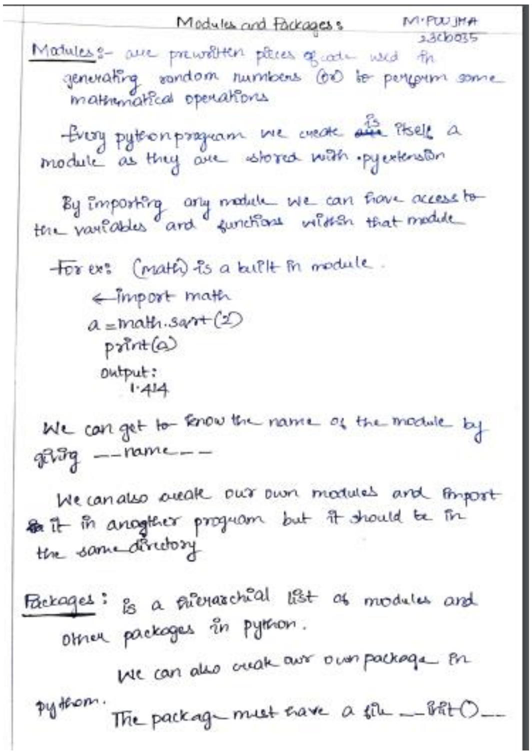
Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO12**	**PSO2**
	3	**1**	**1**	**1**	**2**	**1**
Justification for correlation	Apply the knowledge of engineering fundamentals	Identify and analyze the applications of list, stack and queue concepts in complex engineering problems.	Design solutions for complex engineering problems by understanding the concepts of modules and packages.	Function effectively as an individual by reflecting the ideas on the given topic.	Recognize the need and have the preparation and ability to engage in independent learning.	Motivates Continuous learning to provide the solution to real world business problems.

- **Images / Screenshot of the practice:**


M. Suriya prakash
Modules and packages
* modules are prewritten pieces of code.
* modules can be definition contain many variables and
* You will create your own module.
* dir () function in return package.
* These are modules are perform common task and numbers, performing mathematical operations etc....
* names space is container
* To cater such situation, namespace is keyword.
* Import is used for import the module and everything.


Modules and packages
M.Manojkumar
9536 2324 4024
Modules are file with extension ".py"
It stores pre defined functions ready to use
There are many inbuilt modules.
Ex:
Math, etc.
we can also create a module and stored it in the same directory as python

Packages
Packages are structured with related function modules
We can create packages by creating a file "__init__.py" in that folder.
Ex: pygame is a package
we can install pygame by using
pip install pygame



Modules and Packages: M.POOJHA 23CB035

Modules:- are prewritten pieces of code used in generating random numbers (or) to perform some mathematical operations

Every python program we create ~~are~~ is itself a module as they are stored with .py extension

By importing any module we can have access to the variables and functions within that module

For ex: (math) is a built in module.

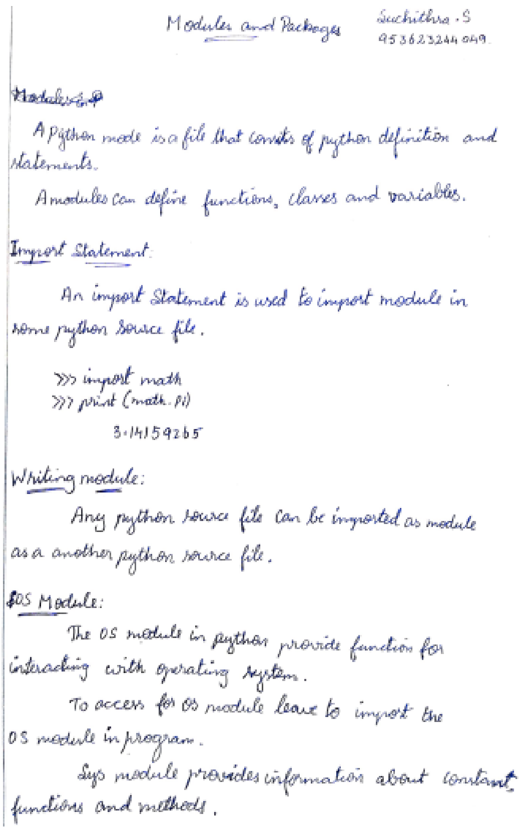
```
import math
a = math.sqrt(2)
print(a)
output:
1.414
```

We can get to know the name of the module by giving __name__

We can also create our own modules and import it in another program but it should be in the same directory

Packages: is a hierarchical list of modules and other packages in python.

We can also create our own package in python. The package must have a file __init()__



Modules and Packages Suchithra. S 953623244049.

Modules:

A python mode is a file that consists of python definition and statements.

A modules can define functions, classes and variables.

Import Statement:

An import statement is used to import module in some python source file.

```
>>> import math
>>> print (math.pi)
3.14159265
```

Writing module:

Any python source file can be imported as module as a another python source file.

OS Module:

The OS module in python provide function for interacting with operating system.

To access for OS module have to import the OS module in program.

Sys module provides information about constant, functions and methods.

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    Students felt that they can able to recall the concepts of modules and packages. And this activity helped them to analyze their level of focus on the classroom.
  - ***Benefit of the practice:***
    1. Students were actively participated in this activity.
    2. It increases students focus in class and also improves the student's writing skill and analyzing ability on the topic.
  - ***Challenges faced in implementation:***

    Some students felt difficult to recall the topics on their own.

References:

1. https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html
2. https://www.unl.edu/gradstudies/current/teaching/minute
3. https://oncourseworkshop.com/self-awareness/one-minute-paper/